Proofs of eleven congruence conjectures of Hanna from the OEIS, and the resolution of three more*

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Abstract

Many entries of the On-Line Encyclopedia of Integer Sequences contributed by Paul D. Hanna define a power series by a functional equation and conjecture a congruence for its coefficients. We prove eleven such conjectures:

- the modulo-three conjecture of A389472;
- the parity conjectures of A120566, A240998, A273958, A295762, A301933, A338633, A338634 and A388734;
- the modulo-three conjectures of A274479 and A377100.

We also resolve three more:

- for A184894 and A107099 we prove the conjectured vanishing modulo 3 outside the exceptional indices. We also show that the stronger reading, non-vanishing at every exceptional index, is false. It fails at the OEIS index 365 of A184894, and at the coefficient of x^{729} (OEIS index 364) of A107099;
- for A361047 we prove the intended statement, which the entry states with an index shift.

Several of these results are quick consequences of classical results, which we point out. Modulo 2, the series of A240998, A295762, A301933, A338633 and A338634 reduce, up to an additive constant, to the Catalan series, and that of A273958, multiplied by x, to $C(C(x^2))$, where C is the Catalan series, so these parity patterns follow from the classical parity of the Catalan numbers. The results for A184894 follow quickly from the classical correspondence for linearised polynomials over $\mathbb{F}_3$. For A120566, A274479 and A388734, identities stated in the entries themselves give short proofs. For A107099 and A361047 the reduced series are classical series, identified by a routine uniqueness argument. Only for the modulo-three conjecture of A389472 and for A377100 did we find neither a proof dated before version 1 of this note (26 September 2026) nor a classical derivation; later proofs of both have appeared, see Section 1. The proofs reduce the functional equations modulo a prime and use the Frobenius identity, formal derivatives, explicit solutions with an x-adic uniqueness argument, and Lucas's theorem. Every theorem on a defining equation is stated for every integer power series satisfying it, under the hypotheses given there. All the mathematics of this paper, including the existence and uniqueness of the normalised solutions and their first terms, is formally verified in Lean 4; statements about the OEIS entries and the literature are not. The proofs were found by AI coding agents working under the author's direction; their role is described at the end of the paper. No human mathematician has reviewed this paper.

* *Version 2.* All the mathematics of this paper is now formally verified in Lean 4, including the existence and uniqueness of the normalised solutions, which version 1 asserted without formalisation; see Sections 1 and 7. The theorems are unchanged. Some remarks are made more precise; the changes are listed under Corrections in `ATTEMPTS.md` of the repository. An earlier corrected version (1.2.1) added the connections with classical results and with identities stated in the OEIS entries, which version 1 (released 2026-09-26) did not cite.

1 Introduction

Throughout, $A(x) = \sum_{n\geq 0} a(n)x^n \in \mathbb{Z}[[x]]$, A' is its formal derivative, and for a series h with $h(0) = 0$ we write $A(h)$ or $A \circ h$ for the composition $\sum_n a(n)h^n$. The results are summarised in Table 1; each conjecture is stated in the corresponding entry [4]. The theorems hold for every integer solution of the defining equation, under the hypotheses stated in each theorem.

With the normalisations of Table 2, the equation of each of the thirteen series entries has exactly one solution in $\mathbb{Z}[[x]]$; for A273958 this is one pair (C, A), and for A338633 and A338634 one series A admitting tails as in Theorem 5(4). The equation of A389472 does not determine $a(2)$ ($A = x$ is also a solution: both sides are then $x^2 + x^3$), and the entry takes $a(2) = 1$; the equation of A240998 forces $a(0) \in \{0, 1\}$ (the constant terms give $a(0)^2 = a(0)$), and the entry takes $a(0) = 1$. In the Lean proofs, integrality for A274479 comes from Lemma 1 modulo 2, and for A107099 from writing $A = x + 2B$. For the continued fractions, the tails exist as x-adic limits. So the theorems are not vacuous, and they apply to these solutions. Table 2 also lists the first terms of each solution.

Existence, uniqueness and these first terms are verified in Lean. That the solution is the sequence of the entry rests on the entry's definition of the sequence by the same equation; no Lean statement mentions the OEIS. The agreement with further terms of the b-files rests on the numerical checks of Section 7, not on Lean.

Entry	Defining equation	Result
A389472	$A(x^2 + x^3) = x^2(1 + A)$	$3 \mid a(3n-1)$, $n > 1$
A240998	$A^2 = x + A(x + 2x^2)$	$a(n)$ odd $\iff$ $n = 2^k$ $(n \geq 1)$
A295762	$A(x - 2A(x^2)) = x + A(x^2)$	$a(n)$ odd $\iff$ $n = 2^k$ $(n \geq 1)$
A273958	$xA + x^2A^2 = C^2$, $C = x + C^2$	$a(n)$ odd $\iff$ $n = 2 \cdot 4^k - 1$
A301933	$A = x(1 + 4AA')/(1 + AA')$	$a(n)$ odd $\iff$ $n = 2^k$ $(n \geq 1)$
A338633	$1 = A - x/(A - 2^3x/(A - 3^3x/\cdots))$	$a(n)$ odd $\iff$ $n = 2^k$ $(n \geq 1)$
A338634	$1 = A - x/(A - 2^4x/(A - 3^4x/\cdots))$	$a(n)$ odd $\iff$ $n = 2^k$ $(n \geq 1)$
A388734	$A = 1 + xA^2 + x^2(1 - x)A^3$	all $a(n)$ odd
A120566	$A = A \circ A - x\, A \circ A \circ A$	all $a(n)$ odd $(n \geq 1)$
A377100	$A = A(x^3)/A(x^2) + A^2$	$a(n) \equiv 1 \pmod 3$ $(n \geq 1)$
A274479	$A^2 = A\big(x^2/(1 - 2x - 4x^2)\big)$	$a(n) \equiv 1 \pmod 3$ $(n \geq 1)$
A361047	$A(x - x^3A'^2) = x$	$[x^m]A \equiv [m = 3^k] \pmod 3$
A184894	iterates of $x + x^3$	vanishing proved; non-vanishing fails at index 365
A107099	$A \circ A = x + 4x^3$	vanishing proved; non-vanishing fails at x^{729}

Table 1: Summary. Normalisations and precise statements are given in the theorems below. Several of these results are quick consequences of classical results or of identities stated in the entries; see "Relation to classical results" below.

Prior work. The modulo-two conjecture of A389472 was proved earlier by Perez Fontelles [6] and, independently, in the `trureturing` repository [16]; those two proofs do not treat the modulo-three conjecture.

Relation to classical results. Several of the conjectures proved here are quick consequences of classical results. Write $C_n = \binom{2n}{n}/(n+1)$ for the Catalan numbers and $C = \sum_{n\geq 1} C_{n-1}x^n$ for the Catalan series, so that $C = x + C^2$. Modulo 2, the series of A295762 and A301933, and those of A240998 (with $a(0) = 1$, as in the entry), A338633 and A338634 minus 1, all equal C; this is how the proof of Theorem 5 proceeds. For A301933 it can also be read off from the remark in the entry that $C = x(1 + 2CC')/(1 + CC')$: since $2 \equiv 4 \equiv 0 \pmod 2$, both equations reduce to $F = x/(1 + FF')$, which, with $F(0) = 0$, determines F. Hence $a(n) \equiv C_{n-1} \pmod 2$ for $n \geq 1$,

Entry	Normalisation	First terms
A389472	$a(2) = 1$	$a(0..6) = 0, 1, 1, 2, 3, 6, 9$
A240998	$a(0) = 1$	$a(0..5) = 1, 1, 1, 2, 11, 86$
A295762	$a(0) = 0$	$a(1..5) = 1, 3, 12, 69, 444$
A301933	$a(0) = 0$	$a(1..5) = 1, 3, 24, 291, 4596$
A338633	none	$a(0..4) = 1, 1, 7, 250, 21867$
A338634	none	$a(0..4) = 1, 1, 15, 1490, 472475$
A273958	none	$a(0..5) = 0, 1, 2, 4, 10, 30$
A388734	none	$a(0..5) = 1, 1, 3, 9, 33, 127$
A377100	$a(0) = 0$, $a(1) = 1$	$a(2..6) = 1, 1, 4, 10, 28$
A274479	$a(0) = 0$, $a(1) = 1$	$a(2..6) = 1, 4, 10, 34, 106$
A120566	$a(0) = 0$, $a(1) = 1$	$a(2..6) = 1, 1, 3, 7, 33$
A361047	$a(0) = 0$	$[x^1, \ldots, x^5, x^7] = 1, 0, 1, 0, 9, 159$
A107099	$a(0) = 0$, $[x^1]A = 1$	$[x^2, x^3, x^5, x^7, x^9] = 0, 2, -6, 36, -266$
A184894	not applicable	$a(1..5) = 1, 2, 9, 78, 1045$

Table 2: Normalisations and first terms of the solutions, in the coefficient indexing of this paper (statements `EX_*` and `DATA_*` of Table 3). In A361047 and A107099 the listed coefficients are those of x^n, not the indexing of the entry. That these values agree with the data of the OEIS was compared outside Lean. A184894 is defined by a recursion, so there is nothing to normalise.

and these five patterns follow from the classical fact that C_n is odd if and only if $n = 2^k - 1$, a corollary of $\nu_2(C_n) = s_2(n+1) - 1$, where ν_2 is the 2-adic valuation and s_2 the binary digit sum [1, Theorem 2.1 in arXiv:math/0407326v1]. Lemma 3 is this fact. The continued fractions of A338633 and A338634 are the cases $e = 3, 4$ of a family whose cases $e = 1, 2$ give, according to those entries, A000699 and A158119. All of them reduce to $A(A-1) = x$ modulo 2, and the parity of A000699 is recorded in A036987 (J. M. Campbell, 2016). For A273958, formula (1) of the entry, which solves the defining equation as a quadratic in xA, gives $xA = -C(-C^2)$, hence $xA \equiv C(C(x^2))$ (mod 2). The congruence $C(C(x)) \equiv \sum_{k\geq 0} x^{4^k}$ (mod 2) is also the key step of the proof of Hanna's conjecture for A374568 in the `trureturing` repository [15]. Parts (1) and (2) of Theorem 12 are an instance of the classical correspondence for linearised polynomials over $\mathbb{F}_3$, and the residues at the exceptional indices with $j \geq 1$ of A184894 and A107099 are, up to sign, Catalan numbers modulo 3; see the remark at the end of Section 6. For A361047 the reduced series coincides with that of the ternary numbers A001764, whose residues modulo 3 are known; see Remark 11. For A120566, A274479 and A388734 we found no classical connection, but identities stated in the entries themselves give short proofs; see the remarks after Theorems 6, 8 and 9.

Apart from the connections with classical results noted above, we found no proof of any of the results in this paper dated before version 1 (26 September 2026) in the OEIS entries and their histories, in the literature, or in the public repositories of proofs of OEIS conjectures that we searched (among them [7, 17]). For the modulo-three conjecture of A389472 and for A377100 we also found no classical connection. This reports the coverage of our searches, not a guarantee of priority.

Later work. After version 1 was released, the following proofs appeared. In the `trureturing` repository, with Lean formalisations: A274479 and A377100 [12] (merged 19 October 2026), the vanishing part of A107099 [13] (merged 19 October 2026), and the modulo-three conjecture of A389472 [14] (merged 6 November 2026). The proof for A389472 there uses the coefficient recurrence, Lucas's theorem and strong induction, and treats the sequence with $a(2) = 1$; our Theorem 4 uses the second derivative and holds for every integer solution. The proofs for A274479 and A377100 there use the same reduction to $x/(1-x)$ modulo 3 as ours. Without formalisation: a report in the `ProveIt` repository [10] (7 October 2026) proves the parity

statement of A338634, and Su [11] (21 October 2026) proves those of A338633 and A338634, in both cases by reducing to $A^2 + A = x$ modulo 2. Our searches on 6 November 2026 found no other work on the fourteen entries, and the OEIS entries were unchanged since version 1. This reports the coverage of our searches, not a guarantee of priority.

2 Tools

Reduction of the coefficients modulo a prime p is a ring homomorphism $\mathbb{Z}[[x]] \to \mathbb{F}_p[[x]]$; it commutes with formal differentiation and with composition by series with zero constant term. We write $F = \overline{A}$ for the reduction and $f(n)$ for its coefficients.

Lemma 1 (Frobenius)**.** *For $F \in \mathbb{F}_p[[x]]$ we have $F(x)^p = F(x^p)$.*

Proof. The map $G \mapsto G^p$ is a continuous ring endomorphism of $\mathbb{F}_p[[x]]$ that fixes $\mathbb{F}_p$ and sends x to x^p. □

Lemma 2. *Let K be a field, let $h \in xK[[x]]$ have order at least two, and let $H \in K[[x]]$ satisfy $H(h) = H$. Then H is constant.*

Proof. Put $B = H - H(0)$, so $B(h) = B$. If $B \neq 0$, its order d is positive, while $\operatorname{ord} B(h) \geq 2d > d$. □

Lemma 3 (dyadic recursion)**.** *Let $F \in \mathbb{F}_2[[x]]$ satisfy $F(0) = 0$ and $F = x + F^2$. Then for $n \geq 1$, $f(n) = 1$ if and only if n is a power of 2.*

Proof. By Lemma 1, $F = x + F(x^2)$, so $f(n) = [n = 1] + [2 \mid n] f(n/2)$ for $n \geq 1$. By strong induction, $f(1) = 1$, $f(n) = 0$ for odd $n \geq 3$, and $f(2m) = f(m)$. □

Since the reduction of the Catalan series C satisfies $F = x + F^2$, Lemma 3 is the classical fact that C_n is odd if and only if $n = 2^k - 1$; see Section 1.

We also use the following observation repeatedly: if $D \in \mathbb{F}_p[[x]]$, $x^{n_0} \mid D$, and $x^n \mid D$ implies $x^{n+1} \mid D$ for all $n \geq n_0$, then $D = 0$.

3 The modulo-three conjecture of A389472

Theorem 4. *If $A(x^2 + x^3) = x^2(1 + A(x))$, then $a(3n - 1) \equiv 0 \pmod 3$ for every $n > 1$.*

Proof. Let $F = \overline{A} \in \mathbb{F}_3[[x]]$ and $h = x^2 + x^3$. Then $F(h) = x^2(1 + F)$ and $h' = 2x + 3x^2 = -x$. Differentiating, $-xF'(h) = -x(1 + F) + x^2F'$, so $F'(h) = 1 + F - xF'$ after cancelling $-x$. Differentiating again, the terms F' cancel and $-xF''(h) = -xF''$, so $F''(h) = F''$. By Lemma 2, F'' is constant. For $n > 1$ the coefficient of x^{3n-3} in F'' is therefore zero, and it equals $(3n - 1)(3n - 2)f(3n - 1) = 2f(3n - 1)$. □

4 Parity via the dyadic recursion

Theorem 5. *In each of the following cases, $a(n)$ is odd for $n \geq 1$ if and only if n is a power of 2:*

1. *(A240998) $A^2 = x + A(x + 2x^2)$;*
2. *(A295762) $a(0) = 0$ and $A(x - 2A(x^2)) = x + A(x^2)$;*
3. *(A301933) $a(0) = 0$ and $A(1 + AA') = x(1 + 4AA')$;*

4. *(A338633, A338634) A satisfies the continued fraction $1 = A - x/(A - 2^e x/(A - 3^e x/(A - \cdots)))$ with $e = 3$, resp. $e = 4$, in the following sense: there are series $T_0, T_1, \ldots$ with constant terms ± 1 such that $(A-1)T_0 = x$ and $(A - T_k)T_{k+1} = (k+2)^e x$ for all $k \geq 0$.*

Moreover (A273958), if $C = x + C^2$ and $xA + x^2A^2 = C^2$, then $a(n)$ is odd if and only if $n = 2 \cdot 4^k - 1$ for some $k \geq 0$.

Proof. (1) Modulo 2, $(x+2x^2)^k \equiv x^k$, so $F^2 = x+F$. Then $G = F - f(0)$ satisfies $G(0) = 0$ and $G^2 = F^2 - f(0) = x+G$, and Lemma 3 applies to G. (2) Modulo 2 the inner series reduces to x, so $F = x + F(x^2) = x + F^2$. (3) Modulo 2 the equation reads $F + F^2F' = x$. In characteristic 2, $F'' = 0$ because $n(n-1)$ is even. Differentiating gives $F' = 1$, hence $F = x + F^2$. (4) Modulo 2, $(A - T_0)T_1 = 0$ with T_1 a unit, so $T_0 = A$ and $(A-1)A = x$. Then $B = A - 1$ satisfies $B(0) = 0$ and $B + B^2 = x$, so $B = x + B^2$, and $a(n) = b(n)$ for $n \geq 1$.

For A273958, constant terms give $C(0)^2 = 0$. By Lemma 3, $c = \overline{C}$ is the indicator of the powers of 2. Put $B = x\overline{A}$, so that $b(n+1) = f(n)$ and $B + B^2 = c^2$, i.e. $B(x) + B(x^2) = c(x^2)$. Hence $b(n) = 0$ for odd n and $b(2m) = b(m) + c(m)$. The powers of 2 split into the disjoint sets $\{4^j\}$ and $\{2 \cdot 4^k\}$, and strong induction gives $b(n) = 1$ exactly for $n = 2 \cdot 4^k$. □

5 Explicit solutions modulo a prime

In the next results we exhibit the reduction F explicitly and show that it is the only solution by the x-adic observation of Section 2.

Theorem 6 (A388734)**.** *If $A = 1 + xA^2 + x^2(1-x)A^3$, then every $a(n)$ is odd.*

Proof. Modulo 2, $U = 1/(1-x)$ is a solution: $1 + (x + x^2)/(1-x)^2 = 1/(1-x)$ because $x + x^2 \equiv x(1-x)$. Subtracting the equations for F and U gives $F - U = (F - U)\, x\, [F + U + x(1-x)(F^2 + FU + U^2)]$, so $x^n \mid F - U$ implies $x^{n+1} \mid F - U$. Hence $F = U$. □

Remark. Formula (4) of the entry, read as a formula for xA as in the program of the entry, gives another short proof: xA is the series reversion of $\varphi = x(1 - x - x^2)/(1 - x^3)$. (This holds because $B = (1 - x^3)/(1 - x - x^2)$ satisfies $B = 1 + \varphi B^2 + \varphi^2(1 - \varphi)B^3$ with $\varphi = x/B$; since φ has no constant term, the equation $Y = 1 + \varphi Y^2 + \varphi^2(1 - \varphi)Y^3$ has a unique power-series solution, so $A(\varphi) = B$, which is formula (5) of the entry.) Since $1 - x^3 = (1 - x)(1 + x + x^2)$, we have $\varphi \equiv x/(1 + x) \pmod 2$, whose reversion is $x/(1 - x)$; hence $A \equiv 1/(1 - x)$.

Theorem 7 (A377100)**.** *If $a(0) = 0$, $a(1) = 1$ and $A = A(x^3)/A(x^2) + A^2$, then $a(n) \equiv 1 \pmod 3$ for $n \geq 1$.*

Proof. Modulo 3 the equation reads $F(x^2)(F - F^2) = F(x^3)$. The series $E = x/(1 - x)$ is a solution: both sides equal $x^3/(1 - x)^3$, because $(1 - x)^3 = 1 - x^3$ and $1 - 2x \equiv 1 + x$. For $D = F - E$ one computes $E(x^2)\, D\, (1 - F - E) = D(x^3) - D(x^2)(F - F^2)$. If $x^n \mid D$ with $n \geq 2$, the right-hand side is divisible by x^{2n+1}; since $E(x^2)$ is x^2 times a unit and $1 - F - E$ is a unit, $x^{2n-1} \mid D$. As $x^2 \mid D$, we get $F = E$. □

Theorem 8 (A274479)**.** *If $a(0) = 0$, $a(1) = 1$ and $A^2 = A\big(x^2/(1 - 2x - 4x^2)\big)$, then $a(n) \equiv 1 \pmod 3$ for $n \geq 1$.*

Proof. Modulo 3, $h = x^2/(1 + x - x^2)$ and $E = x/(1 - x)$ satisfies $E(h) = x^2/(1 + x - 2x^2) = x^2/(1 - x)^2 = E^2$. For $D = F - E$ we have $D(F + E) = D(h)$. If $x^n \mid D$ then $x^{2n} \mid D(h)$, and $F + E = xW$ with $W(0) = 2$ a unit, so $x^{2n-1} \mid D$. As $x^2 \mid D$, we get $F = E$. □

Remark. The entry already notes, in its comment (1), that $x/(1 - x)$ satisfies $F^2 = F\big(x^2/(1 - 2x + 2x^2)\big)$. As $-4 \equiv 2 \pmod 3$, this is the reduced equation, so with this identity the conjecture has a short proof: only the uniqueness argument above is needed.

Theorem 9 (A120566)**.** *If $a(0) = 0$, $a(1) = 1$ and $A = A \circ A - x\, A \circ A \circ A$, then $a(n)$ is odd for every $n \geq 1$.*

Proof. Modulo 2, $E = x/(1-x)$ satisfies $E \circ E = x/(1-2x) = x$, so $E \circ E - xE \circ E \circ E = x - xE = E$. Let $D = F - E$ and suppose $x^n \mid D$ with $n \geq 2$. Since composition preserves congruences modulo x^n and $F = x + O(x^2)$, we have $D \circ F \equiv D$, $E \circ F - E \circ E = D/((1-F)(1-E)) \equiv D$, and $x(F \circ F \circ F - E \circ E \circ E) \equiv 0$, all modulo x^{n+1}. Since $F \circ F - E \circ E = D \circ F + (E \circ F - E \circ E)$, subtracting the equations for F and E gives $D \equiv 2D \equiv 0 \pmod{x^{n+1}}$. As $x^2 \mid D$, we get $F = E$. □

Remark. Formula (2) of the entry, $A(-A(-x)) = x$, gives a shorter proof. It follows from the defining equation: both $-A(-x)$ and any compositional inverse of A with zero constant term satisfy $B = B \circ B + x\, B \circ B \circ B$, and this equation has only one solution $B = x + O(x^2)$. Modulo 2 we have $-A(-x) \equiv A$, so formula (2) gives $F \circ F = x$, and the defining equation becomes $F = x - xF$, that is, $F = x/(1+x)$.

Theorem 10 (A361047)**.** *If $A(0) = 0$ and $A(x - x^3 A'^2) = x$, then $[x^m]A \equiv 1 \pmod 3$ when m is a power of 3, and $[x^m]A \equiv 0 \pmod 3$ otherwise.*

Proof. Modulo 3 let $S = \sum_{k \geq 0} x^{3^k}$. Then $S' = 1$ and, by Lemma 1, $S - S^3 = S - S(x^3) = x$, so $g \circ S = x$ for $g = x - x^3$. Hence $S \circ g = x$, i.e. $S(x - x^3 S'^2) = x$. For $D = F - S$ with $x^n \mid D$, the inner series $g_F = x - x^3 F'^2$ and $g_S = x - x^3 S'^2$ differ by $x^3 D'(F' + S')$, which is divisible by x^{n+2}. Therefore $0 = F(g_F) - S(g_S) = D(g_F) + \big(S(g_F) - S(g_S)\big) \equiv D \pmod{x^{n+1}}$. Hence $F = S$. □

Remark 11. The entry A361047 writes $A = \sum_{n \geq 1} a(n) x^{2n-1}$ and conjectures $a(n) \equiv 1 \pmod 3$ exactly for $n = (3^k - 1)/2$. With this indexing the statement is contradicted by the data $(a(2) = 1)$. It is correct for the index $i = n - 1$, since x^{2i+1} with $i = (3^k - 1)/2$ is x^{3^k}. Theorem 10 contains this intended statement.

Over $\mathbb{Z}$ the reversion of $x - x^3$ is $V = \sum_{k \geq 0} t(k) x^{2k+1}$, where $t(k) = \binom{3k}{k}/(2k+1)$ are the ternary numbers (A001764), since $V - V^3 = x$. Modulo 3 this reversion is S, so Theorem 10 says that $a(n) \equiv t(n-1) \pmod 3$. The residues of $t(k)$ modulo 3 form A113047, which is 1 for $k = (3^j - 1)/2$ and 0 otherwise; this was conjectured by P. Barry and proved by K. Ryde in 2021 [9]. The statement in A361047 is this pattern without the index shift. The remaining step, $F = S$, or equivalently $F' = 1$, which reduces the defining equation modulo 3 to $F(x - x^3) = x$, follows by the routine uniqueness argument in the proof above.

6 Additive series modulo three

Call $G \in \mathbb{F}_3[[x]]$ *additive* if $[x^m]G = 0$ whenever m is not a power of 3. By Lemma 1, $G^{3^j} = G(x^{3^j})$, so the composition of two additive series is additive.

Theorem 12 (A184894)**.** *Let $P_0 = x$, $P_{m+1} = P_m + P_m^3$, and $a(m) = [x^{2m-1}]P_m$ for $m \geq 1$.*

1. *If $2m - 1$ is not a power of 3, then $3 \mid a(m)$.*
2. *If $2m - 1 = 3^j$, then $a(m) \equiv \binom{m}{j} \pmod 3$.*
3. *$3 \mid a(365)$, although $2 \cdot 365 - 1 = 3^6$.*

Proof. Modulo 3, $P_{m+1} = P_m + P_m(x^3)$, so the coefficients $c(m,k) = [x^k]P_m$ satisfy $c(m+1,k) = c(m,k) + [3 \mid k]\, c(m, k/3)$ with $c(0,k) = [k = 1]$. By induction, $c(m,k) = 0$ unless k is a power of 3, and $c(m, 3^j) = \binom{m}{j}$. Finally $\binom{365}{6} \equiv 0 \pmod 3$ by Lucas's theorem, since $365 = (111112)_3$ and $6 = (20)_3$. □

The entry conjectures that $a(m) \equiv 0 \pmod 3$ except at $m = (3^n + 1)/2$. Part (1) proves this statement as written, namely vanishing outside the exceptional indices. Part (3) shows that the stronger reading, that $a(m)$ is nonzero modulo 3 at every exceptional index, is false; the first failure is at $n = 6$. By Lucas's theorem, $a((3^n + 1)/2) \not\equiv 0$ exactly when every base-3 digit of n except the units digit is at most 1.

Theorem 13 (A107099)**.** *Let* $A(0) = 0$, $[x^1]A = 1$ *and* $A \circ A = x + 4x^3$. *Then* $3 \mid [x^n]A$ *whenever* n *is not a power of* 3, *and* $3 \mid [x^{729}]A$.

Proof. Modulo 3 the equation reads $F \circ F = x + x^3$. Suppose m is not a power of 3 and $f(j) = 0$ for all non-powers $j < m$. Let T be the truncation of F below degree m; it is additive, so $[x^m]T \circ T = 0$. Since $F = T + f(m)x^m + O(x^{m+1})$ and $f(1) = 1$, we get $[x^m]F \circ F = 2f(m)$, while $[x^m](x + x^3) = 0$; hence $f(m) = 0$. Thus $F = \sum_k c_k x^{3^k}$ is additive, and comparing coefficients of x^{3^k} in $F \circ F = \sum_i c_i F(x^{3^i})$ gives $c_0 = 1$ and $2c_k + \sum_{0<i<k} c_i c_{k-i} = [k = 1]$ for $k \geq 1$. This yields $(c_1, \ldots, c_6) = (2, 1, 1, 2, 1, 0)$. □

The entry writes $A = \sum_{n\geq 0} a(n)x^{2n+1}$ and conjectures that $[x^n]A \equiv 0 \pmod 3$ except at $n = 3^k$. Theorem 13 proves this statement as written. It also shows that the stronger reading, non-vanishing at every power of 3, is false: $[x^{729}]A = a(364) \equiv 0 \pmod 3$. The OEIS b-file of A107099 already shows $a(364) \equiv 0 \pmod 3$.

Remark. Parts (1) and (2) of Theorem 12 are an instance of the classical correspondence between linearised polynomials over $\mathbb{F}_3$ and their conventional 3-associates, $\sum_j c_j x^{3^j} \leftrightarrow \sum_j c_j t^j$, under which composition corresponds to multiplication [5], [2, Lemma 3.59]: since $x + x^3$ corresponds to $1+t$, we get $P_m \equiv \sum_j \binom{m}{j} x^{3^j} \pmod 3$. Likewise, the correspondence, which extends to series, shows that the additive series $\sum_j \binom{1/2}{j} x^{3^j}$, which corresponds to $(1+t)^{1/2}$, composes with itself to $x + x^3$ modulo 3. By a routine uniqueness argument the reduction F in Theorem 13 is this series. For $j \geq 1$ we have $\binom{1/2}{j} = 2(-1)^{j-1}C_{j-1}/4^j \equiv (-1)^j C_{j-1} \pmod 3$, and $\binom{(3^j+1)/2}{j} \equiv \binom{1/2}{j} \pmod 3$, because modulo 3 the series $(1 + t)^{(3^j+1)/2}$ and $(1 + t)^{1/2}$ both have constant term 1 and both square to $1 + t$ modulo t^{3^j}, so they agree below degree 3^j, and $j < 3^j$. So in both entries the residue at an exceptional index with $j \geq 1$ is $(-1)^j C_{j-1}$ modulo 3. By [1, Theorem 5.2 in arXiv:math/0407326v1], it is nonzero exactly when every base-3 digit of j except the units digit is 0 or 1. Both failures come from $3 \mid C_5 = 42$.

7 Verification

Numerical checks. For each entry, one of two checks was made. Either a prefix of the sequence was recomputed from the defining equation and compared with the b-file of the entry, or the b-file was substituted into the defining equation and checked modulo two primes. The lengths and moduli vary between entries and are recorded with the verification programs [8]. All checks agree with the theorems. They are cross-checks only, and they are not Lean statements. Beyond the terms of Table 2, they are the only comparison with the data of the entries.

Formal verification. All the mathematics of this paper is formally verified in Lean 4 (version 4.34.1) with Mathlib [3] (commit `d13f23b7`): each claim is one of the 84 statements of Table 3, a definition, a step of a written proof of one of them, or a statement about a cited source or an OEIS entry, which is marked as such. The statements are in `lean/Research/Backfill/Paper 2/Challenge.lean` of the repository [8], the proofs in the same folder, and `Check.lean` there closes each statement with its proof; the logs are in the repository too. Tests in the repository confirm that `A.subst h` denotes the composition $A(h)$. For example, Theorem 5 for A273958 is, in the form of the statement `Sec7_quotedA273958`, in ASCII notation,

```
forall (Cs A : PowerSeries Int), Cs = X + Cs ^ 2 ->
  X * A + X ^ 2 * A ^ 2 = Cs ^ 2 -> forall n : Nat,
    (Odd (coeff n A) <-> exists k : Nat, n + 1 = 2 * 4 ^ k)
```

Place in the paper	Lean statements
Section 1, existence and uniqueness	`EX_` followed by A389472, A240998, A295762, A301933, A338633, A338634, A273958, A388734, A377100, A274479, A120566, A361047, A107099
Section 1, first terms	`DATA_` followed by the same thirteen entries and A184894
Section 1, classical results	`RC_abstractMod2`, `RC_seriesMod2`, `RC_coeffMod2`, `RC_catalanParity`, `RC_DS21`, `RC_A301933_catalanIdentity`, `RC_A301933_determines`, `RC_CFfamilyMod2`, `RC_A273958_closedForm`, `RC_A273958_mod2`, `RC_CCmod2`
Section 2	`Sec2_reduction`, `Lemma1`, `Lemma2`, `Lemma3`, `ObsXadic`, `CatalanEq`
Theorems 4, 5	`Thm4`, `Thm5_1`, `Thm5_2`, `Thm5_3`, `Thm5_4_A338633`, `Thm5_4_A338634`, `Thm5_A273958`
Theorems 6–10 and remarks	`Thm6`, `Rem6`, `Thm7`, `Thm8`, `Rem8`, `Thm9`, `Rem9`, `Thm10`
Remark 11	`Rem11_intended`, `Rem11_literalFalse`, `Rem11_ternaryReversion`, `Rem11_mod3`, `Rem11_A113047`, `Rem11_remainingStep`
Section 6	`Sec6_additiveComp`, `Thm12_1`, `Thm12_2`, `Thm12_3`, `Thm12_entryLiteral`, `Thm12_firstFailure`, `Thm12_digits`, `Thm13_1`, `Thm13_2`, `Thm13_values`
Remark on additive series	`Rem13_correspondence`, `Rem13_PmMod3`, `Rem13_sqrtSeries`, `Rem13_binomHalf`, `Rem13_lucasHalf`, `Rem13_residues`, `Rem13_DS52`, `Rem13_C5`
Section 7	`Sec7_quotedA273958`

Table 3: The 84 Lean statements, by place in the paper.

What the formalisation does not claim. The Lean statements say nothing about priority, about the contents of the OEIS entries or of other repositories, or about the numerical checks. They do not identify the series in Lean with the infinite sequences of the entries (Section 1), and they do not state that the entry's finite truncations of a continued fraction converge to the tails of Theorem 5(4). The attributions to Campbell, Barry, Ryde and the `trureturing` repository, and the route through formula (1) of A273958, are not formalised.

Statements before formal proofs. Version 1 had a Lean proof of each theorem, written with the proofs. The statements of version 2 were written later than the paper and its proofs. On 27 September 2026 a Claude Code agent wrote them from version 1.2.1, and an independent Claude Code agent reviewed them (83 accepted, one to be changed). On 5 November 2026 OpenAI Codex ported them to Lean 4.34.1 and made the change, to `Rem11_remainingStep`, which now states the step $F = S$ of Remark 11. A Codex agent that had not written the files reviewed them, and they were frozen on 5 November 2026 at 21:15 UTC with 43 tests and 7 regression tests.

Formal proofs. The proofs are in files that import the statement file and do not modify it. On 5 and 6 November 2026 Codex proved the 84 statements in ten groups; each group was accepted after builds in two new empty directories, a check of each module with `leanchecker`, and a review by a Codex agent that had not written it, and the last group was accepted on 6 November 2026 at 03:48 UTC. The builds of the last group rebuilt all 59 modules and gave byte-identical `.olean` files and 332 axiom listings, each within `propext`, `Classical.choice` and `Quot.sound`. No `sorry` and no native evaluation are used. Mathlib and the toolchain were not rebuilt. The statements are restated in `Check.lean`, which Claude Code generated from the list of accepted statements. From 6 to 7 November 2026 the whole Lean project of the repository, with these 59 modules and `Check.lean`, was rebuilt from an empty directory, one module at a time and not with `lake`: all 2901 modules compiled, and 2900 passed `leanchecker`.

The remaining module is the root of the project, which has only imports and no constants, so it was not replayed. The 84 axiom lists printed by `Check.lean` consist of exactly the three axioms above. This part of the repository has not been checked with comparator or with an independent kernel.

Tool and computational resource disclosure

This work was carried out between 25 and 26 September 2026 in a workflow directed by the author, using two AI coding agents: OpenAI Codex (desktop application; models `gpt-6-astra` and `gpt-6-sol`) and Anthropic Claude Code (model Claude Opus 5.5). The author chose the research programme and directed the agents. Codex found and formalised the proof of Theorem 4. Claude Code agents searched the OEIS for candidate conjectures, found and formalised the other proofs, and drafted this text. Each proof was checked by a second agent, and the agents searched the literature for previous proofs. After version 1 was released, a further agent check, which compared the reduced series with classical sequences, found the connections with classical results and with identities stated in the OEIS entries described in Section 1; Claude Code drafted the corresponding changes, which an independent Claude agent checked. Each proof was checked by an AI agent other than the one that found it, and the formal statements were compared with the OEIS entries by a second agent.

Version 2 was produced from 27 September to 7 November 2026. On 27 September 2026 a Claude Code agent (model Claude Opus 5.5) wrote the statements of Section 7, and an independent Claude Code agent (same model) reviewed them; Codex reviewed them on the same day. In November 2026 OpenAI Codex (desktop application; its main session used the model `gpt-6.1-sol`) ported them to the current Lean, wrote all the formal proofs and had them reviewed as described in Section 7. Claude Code (model Claude Opus 5.5) compared version 1.2.1 claim by claim with the statements, drafted the text of version 2 and wrote `Check.lean`. On 6 and 7 November 2026 Claude Code rebuilt the whole Lean project of the repository on the author's Windows computer. Claude Code agents searched the literature again on 6 November 2026. An independent Claude Code agent (model Claude Fable 5.1), which had written none of the files, reviewed a draft of this paper against the statements and the evidence on 6 November 2026, and before the release a second such agent (same model) reviewed the paper, the Lean package and the receipts; neither ran Lean. The corrections of both reviews were applied. Throughout, "we" refers to this workflow; the reading and searching described in the paper were done by the agents. The author is not a professional mathematician. The author has read version 2 of this paper, but did not check its proofs and did not compare its Lean statements with the paper. No human mathematician has reviewed this paper. The correctness of the results rests on the Lean proofs of the statements in Table 3 and on the written proofs; the numerical checks are additional cross-checks. AI systems are not authors of this paper; the author takes full responsibility for its content.

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